Interpretable by Design, Efficient by Distillation: An Industrial Study of Expert-Guided Search-Based Software Team Formation [Supplementary Material]

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I. Introduction

This document provides supplementary material for the paper *Interpretable by Design, Efficient by Distillation: An Industrial Study of Expert-Guided Search-Based Software Team Formation*. It preserves detailed experimental evidence summarized in the main paper without altering or recalculating any data.

Table S1 characterizes the industrial dataset used to calibrate and evaluate the pipeline, including the candidate pool, technical taxonomy, temporal project split, team-size distribution, and collaboration graph. Table S2 documents the retained practitioner-validation design and distinguishes unique cases, repeated comparisons used to examine consistency, and isolated acceptability judgements.

Table S3 provides the complete project-level evidence for RQ2 by comparing the exact MILP reference and the calibrated GA through solution quality, stochastic variability, optimum-hit counts, runtime, and speedup. Table S4 provides the complete project-level evidence for RQ3 by comparing the calibrated-GA recommendations with the historical teams through overall team fit, collaborative adequacy, and relative gain on the sixteen held-out 2025 projects.

Together, these tables support reproducibility and allow readers to trace the aggregate results and interpretations reported in the paper to their underlying project-level evidence.

TABLE S1: Summary of the industrial dataset.

Aspect	Summary
Candidate pool	512 developers with at least one classified technical competence.
Technical taxonomy	13 domains, 213 ecosystems, and 23 programming languages. Median per developer: 2 domains, 6 ecosystems, and 3 languages.
Calibration projects	205 projects starting up to 2024. Team sizes: k=4: 45, k=5: 38, k=6: 37, k=7: 30, k=8: 26, k=9: 29.
Held-out projects	16 projects starting in 2025. Team sizes: k=4: 4, k=5: 4, k=6: 1, k=7: 1, k=8: 2, k=9: 4.
Collaboration graph	Pairwise collaboration history extracted from prior project co-participation and transformed through the expert-calibrated compatibility function.

TABLE S2: Retained practitioner-validation design.

Task	Unique cases	Repeats	Rows
Calibrated GA vs. historical	16	3	19
Calibrated GA vs. uniform GA	12	2	14
Calibrated-GA acceptability	16	–	16

TABLE S3: RQ2 comparison between the exact MILP reference and the calibrated GA on the twelve size-four controlled benchmark projects.

Project	MILP AE	GA best AE	Best gap (%)	GA AE mean $\pm$ SD	Hits/30	MILP time (s)	Speedup
P1	0.818783	0.818783	0.00	0.788474 $\pm$ 0.050642	12	109.69	7.85$\times$
P2	0.844445	0.844445	0.00	0.769284 $\pm$ 0.059689	7	208.44	10.26$\times$
P3	0.813492	0.810847	0.33	0.789697 $\pm$ 0.032128	0	186.75	7.61$\times$
P4	0.864286	0.864286	0.00	0.817566 $\pm$ 0.033847	1	45.74	4.41$\times$
P5	0.843915	0.826720	2.04	0.771497 $\pm$ 0.032369	0	34.46	2.20$\times$
P6	0.781746	0.781746	0.00	0.753527 $\pm$ 0.035620	14	508.73	93.51$\times$
P7	0.870900	0.859326	1.33	0.858016 $\pm$ 0.004189	0	585.89	117.37$\times$
P8	0.857672	0.857672	0.00	0.790728 $\pm$ 0.054681	10	195.47	27.33$\times$
P9	0.867725	0.867725	0.00	0.854559 $\pm$ 0.023670	16	122.19	15.06$\times$
P10	0.900794	0.900794	0.00	0.876323 $\pm$ 0.024483	11	105.33	26.60$\times$
P11	0.834656	0.834656	0.00	0.779374 $\pm$ 0.031629	2	73.92	12.84$\times$
P12	0.877513	0.857672	2.26	0.794899 $\pm$ 0.028884	0	226.69	45.79$\times$
Summary	–	–	0.50 mean	–	73/360	–	13.95$\times$ median

TABLE S4: RQ3 backtest on the sixteen held-out 2025 projects.

Project	k	$\mathrm{AE}_{\mathrm{real}}$	$\mathrm{AE}_{\mathrm{GA}}$	Gain (%)	$\mathrm{AC}_{\mathrm{real}}$	$\mathrm{AC}_{\mathrm{GA}}$
P1	9	0.5238	0.6114	16.72	0.5143	0.8111
P2	8	0.3750	0.5167	37.78	0.3643	0.6000
P3	5	0.4532	0.6822	50.53	0.4200	0.7200
P4	9	0.4008	0.6506	62.32	0.3571	0.8167
P5	4	0.5762	0.6365	10.47	0.6000	0.8667
P6	7	0.5246	0.5393	2.81	0.7667	0.7952
P7	5	0.5817	0.6326	8.75	0.6333	0.8600
P8	5	0.4365	0.6757	54.80	0.4000	0.8400
P9	6	0.4048	0.5311	31.20	0.4000	0.6467
P10	8	0.5869	0.6162	5.00	0.6643	0.8429
P11	9	0.3274	0.5041	53.98	0.3071	0.6833
P12	4	0.3770	0.5178	37.35	0.3667	0.7667
P13	4	0.7817	0.8643	10.56	0.7667	0.9000
P14	4	0.4008	0.9008	124.75	0.3000	0.9000
P15	5	0.3770	0.5347	41.84	0.3667	0.7400
P16	9	0.4373	0.4963	13.48	0.4810	0.7667
Mean	–	0.4728	0.6194	35.15	0.4818	0.7847